LESSON 9.2

# Analyze Graphs of Relationships Between Two Variables

**Start here** · Words, worked example, and sentence starters

**Name** _____ **Date** _____ **Period** _____

**LEARNING TARGET** I can use a table and a graph to determine and analyze the relationship between two variable quantities.

**WHY THIS MATTERS** The ticket shows the cost to attend a college football game. How does the total cost of tickets relate to the number of tickets purchased?

START HERE

## Words for this lesson

Read each word before you begin. Say it out loud.

| WORD | WHAT IT MEANS | EXAMPLE |
|---|---|---|
| **graph** *Spanish: gráfica* | A picture on a coordinate plane that shows how two quantities are related. | points that rise in a line |
| **table of values** *Spanish: tabla de valores* | A list of matched pairs of values that shows a relationship using numbers. | months 1, 2, 4 beside cost 24.95, 49.90, 99.80 — one row per pair. |
| **ordered pair** *Spanish: par ordenado* | Two numbers written as (x, y) that name one point on a graph. The independent variable comes first. | (3, 12) |
| **axis** *Spanish: eje* | One of the two number lines that frame a graph. Each axis is labeled with a quantity from the problem. | x-axis and y-axis |
| **coordinates** *Spanish: coordenadas* | The numbers in an ordered pair that tell how far to move along each axis to reach a point. | x-coordinate, y-coordinate |
| **Rate of change** *Spanish: Tasa de cambio* | How much y changes each time x goes up by one. | Up $4 for each hour. |

HOW IT WORKS

## Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

**A table for y = 4x has x = 1, 2, 3. Plot the points.**

1. Write the pairs: (1, 4), (2, 8), (3, 12).
2. Plot (1, 4): right 1, then up 4. Plot the others the same way.
3. Check: the points line up.

**Answer:** three points in a straight line

NOW YOU TRY

## Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

**The tram travels 1,200 feet every minute. Which quantity is independent — time or distance?**

1. _____
2. _____
3. _____

**Answer:** _____

SAY IT AND WRITE IT

## Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The independent variable is _____ , so it goes on the _____ axis.
- The ordered pair ( _____ , _____ ) means that _____ .
- To plot a point I move _____ first, then _____ .
- As x increases, y _____ , so the points _____ .

**Word bank** graph · ordered pair · coordinates · axis · plot · point · table · input · output · rate · increases · origin

WATCH OUT!

## A common mistake

Reversing the coordinates of a point

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# Analyze Graphs of Relationships Between Two Variables

**Level 0** · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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**LEARNING TARGET** I can use a table and a graph to determine and analyze the relationship between two variable quantities.

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HOW IT WORKS

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**1** **Circle the letter of the best answer. Show how you know.**

Football tickets cost $45 each. What is the total cost of 3 tickets?

(A) $45
(B) $48
(C) $90
(D) $135

**STEPS TO TRY**

1. What is the cost of ONE ticket?
2. Each of the 3 tickets adds another $45.
3. Multiply the price of one ticket by the number of tickets.

SHOW YOUR WORK

**2** **Circle the letter of the best answer. Show how you know.**

Which ordered pair belongs on the ticket graph for 2 tickets?

(A) (2, 90)

(B) (2, 45)

(C) (2, 47)

(D) (90, 2)

**STEPS TO TRY**

1. Which quantity goes first in the ordered pair?
2. Tickets is the independent variable, so it comes first.
3. The second number is 2 × 45.

**SHOW YOUR WORK**

**3** **Circle the letter of the best answer. Show how you know.**

On the ticket graph, what should the two axes be labeled?

(A) Number of tickets and price of one ticket

(B) Number of tickets and total cost

(C) Total cost and $45

(D) Number of games and number of tickets

**STEPS TO TRY**

1. Which TWO quantities change in this situation?
2. One is the choice; the other responds to it.
3. You choose the tickets, and the total cost responds.

**SHOW YOUR WORK**

**4** **Circle the letter of the best answer. Show how you know.**

The Sandia Peak tram travels 1,200 feet each minute. How far has it traveled after 3 minutes?

(A) 1,200 feet
(B) 1,203 feet
(C) 2,400 feet
(D) 3,600 feet

**STEPS TO TRY**

1. How far does the tram go in ONE minute?
2. Each minute adds another 1,200 feet.
3. Multiply 1,200 by the number of minutes.

**SHOW YOUR WORK**

**EXPLAIN YOUR THINKING**

**Find a card you called a match and your partner did not. Pick one ordered pair from it and test it in cost = 45 × tickets — what does the point on the graph tell you that the table does not?**

**SENTENCE STARTER** The ordered pair ___ does / does not match because 45 × ___ = ___.

**3-Point Exit Checklist:** ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

**How did it go?** ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help